

4.2 The 3×3 Matrix Determinant



Core skills you'll master in this section:

- ▶ The determinant of a 3×3 square matrix represents the signed volume of the parallelepiped formed by its column vectors.
- ▶ Use the right-hand rule to determine the sign of the determinant: positive if the column vectors form a right-handed system, negative otherwise.
- ▶ The identity matrix's column vectors form the unit cube, with volume 1, so its determinant is 1.
- ▶ The determinant of a diagonal matrix equals the product of its diagonal elements, corresponding to the volume of a cube or rectangular box.
- ▶ The determinant of an upper or lower triangular matrix equals the product of its diagonal elements; if any main-diagonal element is 0, the determinant is also 0.
- ▶ If the three column vectors of a 3×3 square matrix are coplanar or collinear (linearly dependent), they cannot span a volume, and the determinant is 0.
- ▶ Swapping any two columns or rows reverses the direction of the right-handed system, and the determinant changes sign.

The previous section discussed the determinant of a 2×2 matrix and revealed its relationship to the area of a parallelogram; this section builds on that by “raising the dimension” — let's look at the connection between the determinant of a 3×3 matrix and the parallelepiped.

Before we begin, let's review two important points about determinants:

- 1) The determinant applies only to square matrices; for a non-square matrix, the determinant is not defined, so the concept of a determinant cannot be used when working with a non-square matrix;
- 2) The determinant operation $\det()$ is a mapping that converts an $n \times n$ square matrix into a scalar value; the determinant can reflect certain key properties of a square matrix, such as the area/volume scaling factor and whether the square matrix is invertible.

The content of this section is arranged almost identically to that of the previous section; we recommend reading them side by side, so you can more intuitively understand the geometric meaning of the determinant across different dimensions.

The Determinant of a 3×3 Matrix

Given the following 3×3 matrix $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

Matrix $\mathbf{A}$'s determinant can be obtained by a weighted sum of three 2×2 determinants, specifically

$$\det(\mathbf{A}) = a_{1,1} \begin{vmatrix} a_{2,2} & a_{2,3} \\ a_{3,2} & a_{3,3} \end{vmatrix} - a_{1,2} \begin{vmatrix} a_{2,1} & a_{2,3} \\ a_{3,1} & a_{3,3} \end{vmatrix} + a_{1,3} \begin{vmatrix} a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{vmatrix}$$

Simplifying further, we get

$$\det(\mathbf{A}) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}$$

For now, let's not worry too much about where this formula comes from — just use it as is. How to derive the above result is a topic we'll cover at the end of this chapter.

In this section, we're more concerned with the geometric perspective — how the parallelepiped helps us more intuitively understand the determinant of a 3×3 matrix.

The Volume of a Parallelepiped

First, split the above 3×3 matrix $\mathbf{A}$ into three column vectors

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \Rightarrow \mathbf{a}_1 = \begin{bmatrix} a \\ d \\ g \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} b \\ e \\ h \end{bmatrix}, \mathbf{a}_3 = \begin{bmatrix} c \\ f \\ i \end{bmatrix}$$

Earlier in this book, we introduced that when summing the three vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$, we can construct a parallelepiped, as specifically shown in Figure 1. Here, we'll borrow this same geometric construction to understand the determinant of a 3×3 matrix.

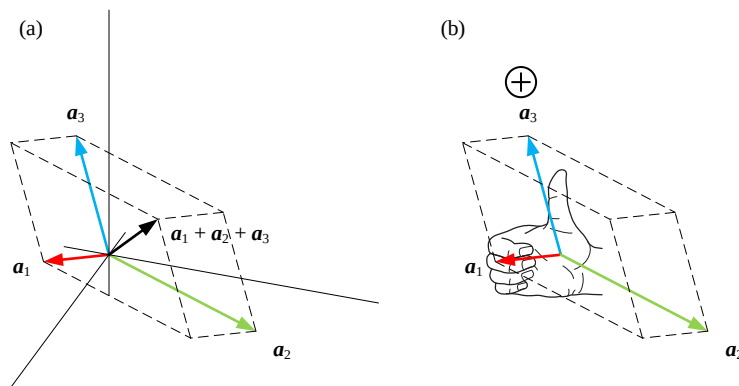


Figure 1. The Parallelepiped

In three-dimensional space, we can use the right-hand rule to determine the sign of a 3×3 matrix's determinant. The specific steps are as follows:

- 1) Make a fist with your right hand, extend your four fingers, and curl them;
- 2) The direction your four fingers curl represents matrix $\mathbf{A}$'s first column vector $\mathbf{a}_1$;
- 3) Your palm faces the second column vector $\mathbf{a}_2$;
- 4) Your extended thumb points toward the third column vector $\mathbf{a}_3$;

If your thumb points in the same direction as the matrix's third column vector $\mathbf{a}_3$, the determinant is positive, and the matrix's column vectors form a right-handed system. Conversely, if the thumb points in the direction opposite to $\mathbf{a}_3$, the determinant is negative, and the matrix's column vectors form a left-handed system.

The Identity Matrix

If the 3×3 square matrix $\mathbf{A}$ is the identity matrix, write $\mathbf{A}$ as three column vectors

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{a}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

As shown in Figure 2, the 3×3 square matrix $\mathbf{A}$'s column vectors form a unit cube.

The identity matrix is clearly a special case of a diagonal matrix; the previous section mentioned that the determinant of a diagonal matrix equals the product of its diagonal elements.

This 3×3 square matrix $\mathbf{A}$'s determinant is

$$\det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1$$

This unit cube's volume is 1.

As shown in Figure 2, the column vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ form the base of this unit cube, whose base area is clearly 1. And $\mathbf{a}_3$ can be regarded as the unit cube's height, that is, 1.

As shown in Figure 2 (b), $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$'s spatial relationship satisfies the right-hand rule, so the determinant is positive.

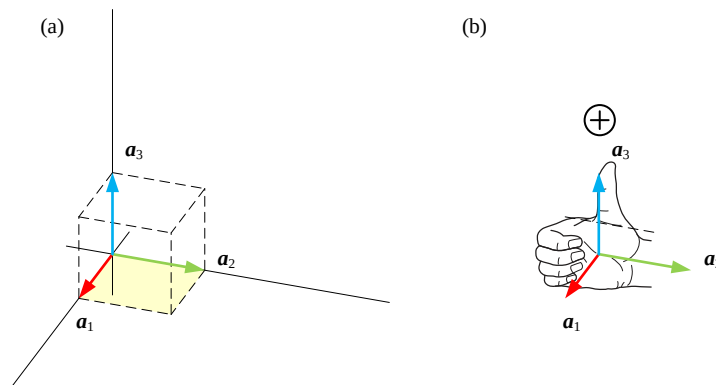


Figure 2. The Unit Cube

Swap 's columns 1 and 2 to get

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \mathbf{a}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{a}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

As shown in Figure 3 (a), the column vectors of the above matrix still span a unit cube, meaning the volume is 1.

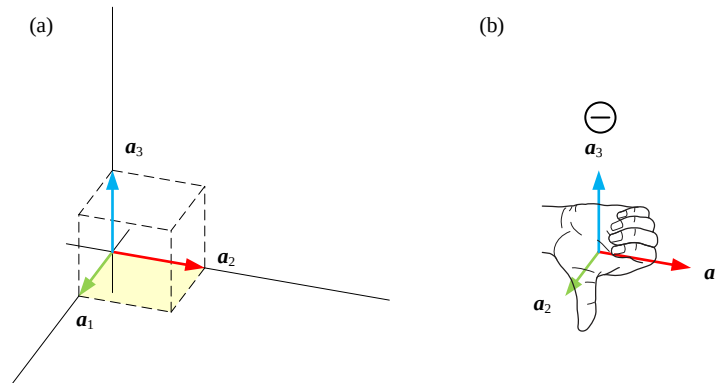


Figure 3. The Unit Cube, with Columns 1 and 2 Swapped

But observing Figure 3 (b), we see that $\mathbf{a}_3$'s direction is opposite to the thumb, so after swapping 's columns 1 and 2, the determinant is negative, that is

$$\det \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = -1$$

Diagonal Square Matrix with Identical Diagonal Elements

The 3×3 square matrix $\mathbf{A}$ is a diagonal square matrix, with identical elements on the diagonal, for example

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \Rightarrow \mathbf{a}_1 = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}, \mathbf{a}_3 = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

The determinant of this square matrix is

$$\det \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = 3 \times 3 \times 3 = 27$$

As shown in Figure 4 (a), 's three column vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ span a cube with side length 3; clearly this cube's volume is 27.

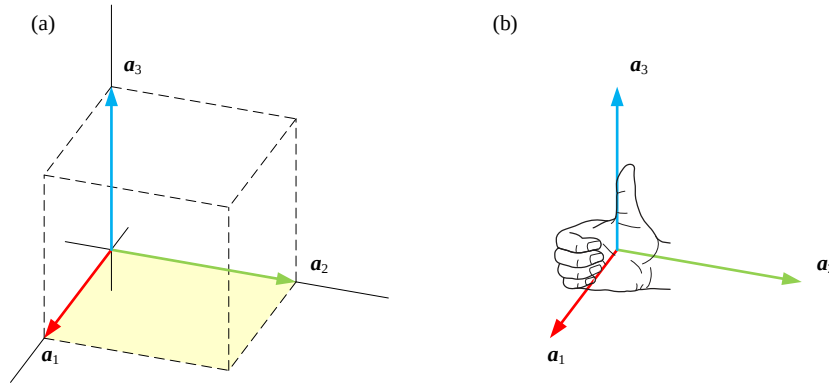


Figure 4. The Cube

Diagonal Square Matrix with Different Diagonal Elements

When the 3×3 square matrix $\mathbf{A}$ is a diagonal square matrix but its diagonal elements differ, $\mathbf{A}$'s corresponding geometric shape is a rectangular box.

As shown in Figure 5 (a), this 3×3 square matrix $\mathbf{A}$'s volume is "length $\times$ width $\times$ height," that is

$$\det \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 6$$

Figure 5 (b), $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ satisfy the right-hand rule, so clearly this determinant is positive.

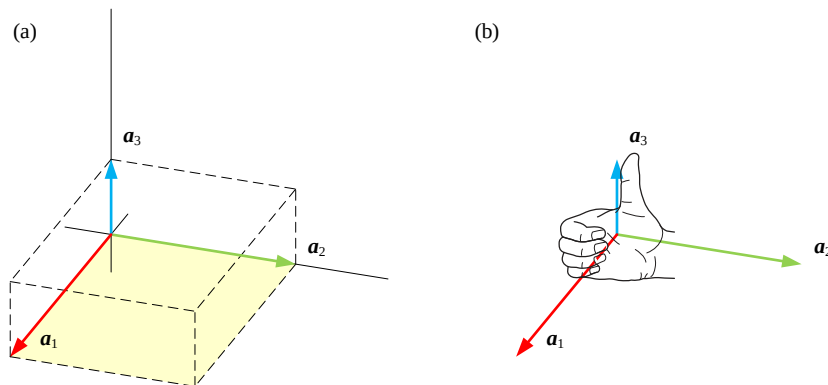


Figure 5. The Rectangular Box

Upper Triangular Matrix

The previous section introduced that the determinant of an upper triangular matrix equals the product of its diagonal elements; let's look at the following example.

Compute the determinant of the following 3×3 upper triangular matrix

$$\det \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1$$

Then, let's look at this 3×3 upper triangular matrix's $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ — three column vectors

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ span the parallelepiped shown in Figure 6 (a).

$\mathbf{a}_1$ is on the x_1 axis; $\mathbf{a}_2$ is on the x_1x_2 plane. $\mathbf{a}_1$ and $\mathbf{a}_2$ form a parallelogram with area 1; this is the parallelepiped's "base."

And this parallelepiped's height is contributed solely by $\mathbf{a}_3$'s x_3 axis component, that is, 1.

It's easy to compute that Figure 6 (a)'s parallelepiped has volume 1, consistent with the result .

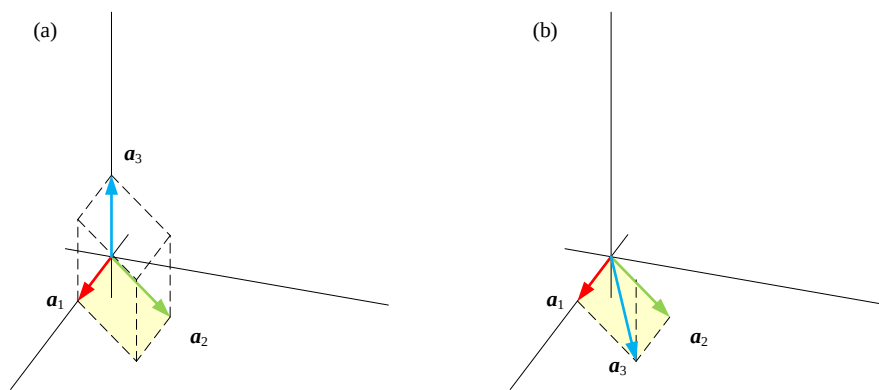


Figure 6. Upper Triangular Matrix

If $\mathbf{a}_1$ and $\mathbf{a}_2$ stay the same, and $\mathbf{a}_3$ is changed to

$$\mathbf{a}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

As shown in Figure 6 (b), we find that $\mathbf{a}_3$ is on the plane spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$. Looked at another way, $\mathbf{a}_3$, $\mathbf{a}_1$, and $\mathbf{a}_2$ — the three vectors are linearly dependent!

At this point, the parallelepiped "lies flat" on the plane, so its volume is 0.

Clearly, the determinant is 0 in this case

$$\det \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

The previous section mentioned that if any diagonal element of an upper triangular matrix is 0, the determinant of that upper triangular matrix is 0.

Lower Triangular Matrix

The previous section mentioned that, similar to the diagonal square matrix and upper triangular matrix, the determinant of a lower triangular matrix equals the product of its diagonal elements.

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The determinant of the following 3×3 lower triangular matrix is 1

$$\det \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = 1$$

This matrix's three column vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ span the parallelepiped shown in Figure 7 (a). Please analyze this parallelepiped's volume on your own.

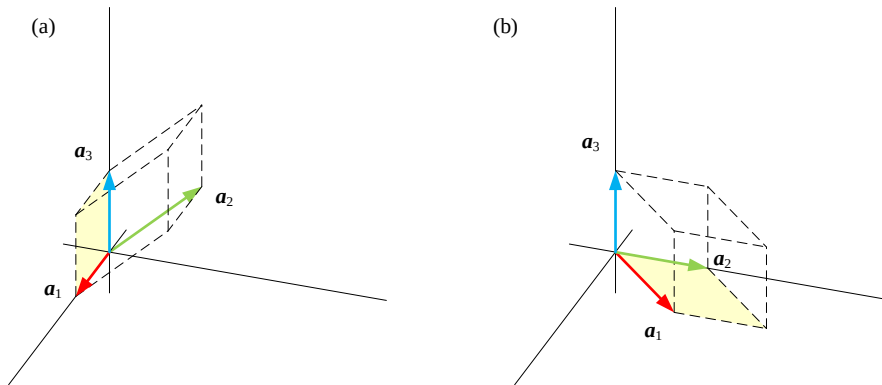


Figure 7. Lower Triangular Matrix

Figure 7 (b) gives a second example of a 3×3 lower triangular matrix, whose corresponding determinant is

$$\det \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1$$

Please also analyze on your own the parallelepiped corresponding to this matrix's determinant.

Code 1 visualizes the parallelepiped corresponding to a 3×3 square matrix. Let's go over the key statements below.

- a** From the `mpl_toolkits.mplot3d.art3d` module, which is dedicated to drawing 3D graphics, we import a class called `Poly3DCollection`; we'll use it later to draw the faces of a 3D solid shape.
- b** Creates a matrix $\mathbf{A}$ with 3 rows and 3 columns.
- c** Extracts the 3 columns from matrix $\mathbf{A}$ and stores them separately in $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$.
- d** Gets the vertices of the parallelepiped.
- e** This list defines six faces, each made up of 4 vertices. Together, these faces form a brick-like solid shape called a parallelepiped.
- f** Creates an empty figure window object `fig`, which will hold all the graphics that follow.
- `fig.add_subplot(111, projection='3d')` adds a subplot to the figure; the parameter 111 means 1 row, 1 column, the 1st subplot, and `projection='3d'` means we're drawing a three-dimensional plot.
- g** `quiver()` draws arrow plots of the column vectors. For example, in `ax.quiver(0, 0, 0, *a1)`, the first 3 numbers are the starting-point coordinates, here 0, 0, 0; `*a1` unpacks $\mathbf{a}_1$'s three coordinates and passes them in as the arrow's direction. Then, the second and third arrows are drawn in the same way.

 Poly3DCollection creates a polyhedron object made up of the six faces defined earlier. `alpha=0.3` sets the transparency, `facecolor='yellow'` makes the faces yellow, `edgecolor='black'` makes the edges black, and `ls='--'` makes the edges dashed.

Then, `ax.add_collection3d(face_collection)` adds the polyhedron we just created to the plot, so it's displayed.

Code 1. Visualizing the Parallelepiped Corresponding to a 3×3 Square Matrix |  LA_04_02_01.ipynb

```

## 初始化
import numpy as np
import matplotlib.pyplot as plt
a from mpl_toolkits.mplot3d.art3d import Poly3DCollection

## 定义一个 3×3 矩阵 A
b A = np.array([[2, 0, 2],
                [2, 2, 0],
                [0, 2, 2]])

## 提取列向量
c a1 = A[:, 0]; a2 = A[:, 1]; a3 = A[:, 2]

## 构造平行六面体的8个顶点
d 0 = np.array([0, 0, 0])
P1 = a1; P2 = a2; P3 = a3
P4 = a1 + a2; P5 = a1 + a3; P6 = a2 + a3
P7 = a1 + a2 + a3
vertices = [0, P1, P2, P4, P3, P5, P6, P7]

## 定义六个面
# 每个面是4个点
e faces = [[0, P1, P4, P2], # 底面
            [0, P2, P6, P3], # 侧面
            [0, P1, P5, P3], # 侧面
            [P7, P4, P1, P5], # 上面
            [P7, P4, P2, P6], # 侧面
            [P7, P5, P3, P6]] # 背面

## 可视化
f fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

# 绘制向量箭头
g ax.quiver(0, 0, 0, *a1, color='red', linewidth=2)
ax.quiver(0, 0, 0, *a2, color='green', linewidth=2)
ax.quiver(0, 0, 0, *a3, color='blue', linewidth=2)

# 绘制平行六面体
h face_collection = Poly3DCollection(faces, alpha=0.3,
                                     facecolor='yellow',
                                     edgecolor='black', ls = '--')
ax.add_collection3d(face_collection)

# 图像装饰
ax.set_xlim([0, 3]); ax.set_ylim([0, 3]); ax.set_zlim([0, 3])
ax.set_xlabel("x1"); ax.set_ylabel("x2"); ax.set_zlabel("x3")
ax.view_init(elev=30, azim=30) # 设置视角
ax.set_proj_type('ortho')
ax.set_box_aspect([1, 1, 1])

```





Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please compute the determinants of the following diagonal square matrices, point out what's special about the parallelepiped formed by the column vectors, and finally use the right-hand rule to determine the sign of the determinant.

▶
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

▶
$$\begin{bmatrix} 1 & & \\ & -1 & \\ & & -1 \end{bmatrix} \text{ (blank entries are 0)}$$

▶
$$\begin{bmatrix} 1 & & \\ & 2 & \\ & & 3 \end{bmatrix}$$

▶
$$\begin{bmatrix} 1 & & \\ & 0 & \\ & & 3 \end{bmatrix}$$

Q2. Please compute the determinants of the following two upper triangular matrices, point out what's special about the parallelepiped formed by the column vectors, and finally use the right-hand rule to determine the sign of the determinant.

▶
$$\begin{bmatrix} 1 & 1 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix}$$

▶
$$\begin{bmatrix} 1 & 1 & 1 \\ & 2 & 1 \\ & & 3 \end{bmatrix}$$

Q3. Please compute the determinants of the following two lower triangular matrices, point out what's special about the parallelepiped formed by the column vectors, and finally use the right-hand rule to determine the sign of the determinant.

▶
$$\begin{bmatrix} 1 & & \\ 1 & 1 & \\ 1 & 1 & 1 \end{bmatrix}$$

▶
$$\begin{bmatrix} 1 & & \\ 1 & 2 & \\ 1 & 1 & 3 \end{bmatrix}$$

Q4. Please swap any pair of column vectors in the matrices from Q1, Q2, and Q3, and recompute the determinant.

